Neddy on your watch board

Part 2, detailed edition: steps 2 to 6 in tiny, one-click pieces. Each small step has a number like **2.4**. If one of them does not match what you see, stop at that number and tell me, for example: “stuck at 4.7, I see …”.

Before you start, have these ready:

- Your Windows PC with the Arduino IDE already set up (Part 1 done and the screen test worked).
- The board and its USB-C **data** cable.
- Your Android phone with the Neddy app, signed in.
- The name and password of your home Wi-Fi (Step 3 shows you how to find them).
- The file **neddy_device.zip** I sent you in our chat.

Step 1 (quick): put the Neddy folder where Arduino can find it

1.1 Find `neddy_device.zip` in your **Downloads** folder (open File Explorer, click **Downloads** on the left).

1.2 Right-click it › **Extract All…** › **Extract**. A normal folder named `neddy_device` opens.

1.3 Click once on the folder named `neddy_device` (the folder itself, not a file inside) and press `Ctrl` + `C` to copy it.

1.4 In File Explorer click **Documents** on the left, double-click the **Arduino** folder, and press `Ctrl` + `V`.

> You should see: `Documents\Arduino\neddy_device`, with the files `neddy_device.ino`, `settings.h` and `es8311.c` inside it. If you see a folder inside a folder with the same name, open the inner one and move *that* out, so `neddy_device.ino` sits directly in `neddy_device`.

A map of the Arduino window (a simplified drawing)

Your colours may differ a little. The numbered red circles are the things these steps refer to.



1	The left icon bar. The **Boards Manager** (circuit board) and **Library Manager** (books) icons are in it. Hover your mouse over an icon to see its name.
2	**Verify** (check mark): builds the software without sending it.
3	**Upload** (arrow): builds it and sends it to the board.
4	The board and port box: it should say **ESP32S3 Dev Module** and a COM number.
5	The file tabs. **settings.h** is the one you edit.
6	The text of the open file.
7	Library Manager icon (stack of books).
8	The black **output panel** at the bottom. It tells you what happened. If something fails, this is what to copy and send me.

Step 2: add the WebSockets add-on

This is a small free program Neddy needs to talk over the internet. You install it once.

2.1 Open **Arduino IDE** (double-click its icon on the desktop). Wait for the window to appear.

2.2 On the **left edge** there is a column of small icons. Hover over them until the label says **Library Manager** (usually the third or fourth, a stack of books, number 7 in the drawing). Click it.

> A panel slides open with a search box at the top.

2.3 Click in the search box and type `WebSockets`. Wait a few seconds for the list to fill.

2.4 Look through the list for the entry that says **WebSockets** and, underneath, **by Markus Sattler**. Its description starts “WebSocket Server and Client for Arduino based on RFC6455”.

> There are several similar names (for example “WebSockets2_Generic” or “ArduinoWebsockets”). Do **not** pick those. You want exactly **WebSockets by Markus Sattler**.

2.5 Click the **INSTALL** button on that entry. If a window asks about installing extra items it needs, click **INSTALL ALL**.

> The button changes to say **INSTALLED**, and the output panel at the bottom says something like “Installed WebSockets”.

2.6 Close the Library Manager panel by clicking its icon again. **Step 2 is done.**

Step 3: type your Wi-Fi into the settings file

3.1 **Find your Wi-Fi name.** On your Windows PC, click the Wi-Fi icon at the bottom right of the screen (near the clock). The network with "Connected" under it is yours. Write its name down **exactly**, including capital letters, spaces and punctuation.

> If you have two names for the same router, such as **Hilton** and **Hilton-5G**, use the one **without** "5G" in it. The board can only use the 2.4 GHz kind. If you have only one name, that is fine: the router will give the board the right kind.

3.2 **Find your Wi-Fi password.** The easiest places:

- **The sticker on the router** (look for "Password", "Key" or "Wi-Fi password").
- **Your Android phone:** Settings › Network & internet (or Connections) › Wi-Fi › tap the little gear or arrow beside your network › **Share**. Your phone shows a QR code and often the password in words under it.
- **Your Windows PC:** press the Windows key, type `Control Panel` and open it › **Network and Internet** › **Network and Sharing Center** › click your Wi-Fi name (blue, next to "Connections") › **Wireless Properties** › **Security** tab › tick **Show characters**.

Passwords are case-sensitive: a capital letter and a small letter are different.

3.3 In the Arduino IDE choose **File › Open...** (or press `Ctrl` + `O`). In the file window click **Documents**, then **Arduino**, then **neddy_device**, click the file `neddy_device.ino` and click **Open**.

> The Arduino window shows a row of tabs: `neddy_device.ino`, `es8311.c`, `es8311.h`, `es8311_reg.h`, `settings.h` (number 5 in the drawing). If it asks to "move the sketch to a folder", click **Cancel** and re-check step 1.4.

3.4 Click the tab named **settings.h**.

> You see a few lines of grey and blue text. Two lines near the top end with `<<< EDIT` in the comment above them:

```
#define WIFI_NAME "your Wi-Fi name"
#define WIFI_PASSWORD "your Wi-Fi password"
```

3.5 **Triple-click** the line that starts `#define WIFI_NAME`. (Click three times fast, and the whole line turns highlighted.) Then **type the whole new line yourself**, replacing it. Here is the exact pattern, using an example network called Hilton Home:

`#define WIFI_NAME "Hilton Home"`

Use **your** name between the quotation marks. The quotation marks are the plain straight ones on the key next to Enter (hold Shift).

3.6 Do the same for the password line: triple-click it, then type, for example:

`#define WIFI_PASSWORD "blue42sky"`

> Do not add anything at the end of the line (no semicolon). If your password contains a quotation mark `"` or a backslash `\`, tell me before going on.

3.7 Press `Ctrl` + `S` to save.

> You should see: the tab name **settings.h** with no dot or asterisk next to it, and the two lines show your real Wi-Fi name and password.

> **If your home Wi-Fi opens a sign-in web page when you connect** (hotels, offices, some shared buildings) the board cannot use it. Use your **phone's hotspot** instead: on Android, Settings › Network & internet › Hotspot & tethering › Wi-Fi hotspot, set a name and a password, turn it on, then type that name and password here.

Step 4: send Neddy to the board (Upload)

4.1 Plug the USB-C cable into the board and into the PC. The board lights up (press the **PWR** button once if it stays dark).

4.2 Look at the box in the toolbar (number 4 in the drawing). Click it. A list drops down: click **Select other board and port...**.

4.3 A window with two columns opens. In the left **BOARDS** search box type `ESP32S3 Dev`. Click **ESP32S3 Dev Module** in the list that appears. In the right **PORTS** column click the entry with **COM** and a number (for example COM5) — the same one that appeared in Part 1.

> If no COM port is listed, try another USB cable or port first. Then do the BOOT-button recovery in step 4.9.

Click **OK**.

4.4 Click the **Tools** menu. You will check six things, one at a time. Each one opens a small list when you hover over it:

Menu item	Pick
USB CDC On Boot	**Enabled**
Flash Size	**32MB (256Mb)** (the 32MB one)
PSRAM	**OPI PSRAM**
Partition Scheme	**Huge APP (3MB No OTA/1MB SPIFFS)**
Upload Speed	leave as it is
Port	your COM number, with a tick

After you click one, the menu closes: click **Tools** again for the next.

4.5 Click the **Upload** button: the round button with an **arrow pointing right** (number 3 in the drawing).

4.6 **Wait.** Nothing else to click. The first time it builds a lot of code, which takes **5 to 10 minutes**. Watch the black output panel at the bottom: lines keep appearing while it works. A small “Compiling sketch…” message pops up at the bottom right, then “Uploading…”.

> When it works the last lines say `Writing at 0x… (100 %)` , then `Hard resetting via RTS pin…` , and a pop-up says **Done uploading**.

4.7 **If red text appears instead** (the first words are usually “Compilation error”): do not change anything. Click inside the black panel, press `Ctrl` + `A` then `Ctrl` + `C` , and paste it into our chat. That is what I need.

4.8 **If it says “Failed to connect to ESP32-S3” or “No serial data received”:** the board is not listening. Do 4.9.

4.9 **Recovery (put the board in download mode):** unplug the USB cable. Press and **hold the BOOT button** (the tiny side button labelled BOOT). While still holding it, plug the cable back in. Keep holding for about 3 seconds, then let go. In the toolbar, pick the COM port again (it may have a new number). Click Upload again.

4.10 When “Done uploading” appears, **unplug the cable and plug it back in** (or press PWR). The board now starts Neddy by itself.

> On the screen, one after another: a green **Neddy** with “Starting”; “Joining Wi-Fi” with your network’s name; then a black screen titled **Pair me with Neddy** with a big six-digit code, for example **482 913**, and a countdown at the bottom.

- Stays on “Joining Wi-Fi” then says **Can’t join Wi-Fi**: the name or password is wrong, or it is not 2.4 GHz. Redo Step 3, then Upload again.
- Says **Can’t reach Neddy**: it has Wi-Fi but cannot reach the server. Check the Wi-Fi really has internet and tell me what number it shows in brackets.
- Screen stays black: unplug and re-plug; check the Tools settings in 4.4.

Step 5: pair the board with your Neddy account

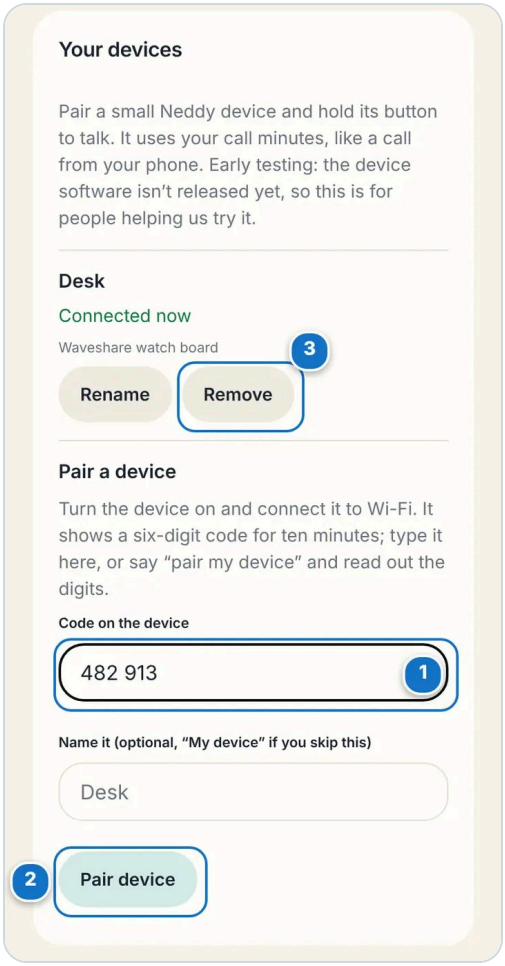
Do this within 10 minutes of the code appearing. If the time runs out the board shows a new code; use the new one.

5.1 Leave the board showing the code. Pick up your **Android phone** and open the Neddy app (the same way you always do) and make sure you are signed in to **your** account.

5.2 At the bottom of the screen, tap **You** (the person icon, bottom right).

5.3 At the top of the You screen you will see tabs or a list of sections. Tap **Account**.

5.4 **Scroll down** past “Appearance”, “Account”, “Passwords” and “Your location”. You will find a card titled **Your devices**, with a smaller heading **Pair a device** and a box labelled **Code on the device**.



This is what that card looks like (a picture from the Help guide). The box number **1** is where you type the code; the button **2** is **Pair device**. Number 3 is the Remove button for a device you already paired.

> No “Your devices” card? That card is shown to the admin. Use the voice way instead: open a Neddy chat and say or type *“Pair my device. The code is 4 8 2 9 1 3.”* with the digits from your board.

5.5 Tap the **Code on the device** box. A number keyboard opens. Type the six digits shown on the board (you can type them with or without a space in the middle). The button **Pair device** turns solid when you have typed all six.

5.6 If you like, tap the next box and type a name, for example `Desk`. (If you skip it the board is called “My device”.)

5.7 Tap **Pair device**.

> **On the phone**, in green: “Paired as … The device should let you know it’s ready in a little while.”
> **On the board**, within a few seconds: **Paired!**, then “Connecting to Neddy”, then a screen with a **green circle**, its name, and **Hold BOOT to talk**.

5.8 If the phone shows a red message: **“That code isn’t right, or it has run out”** — check the digits against the board; if more than 10 minutes passed, type the new code on the board. **“You can pair up to 5 devices”** — tap Remove on one you don’t use. **“…hasn’t matched a few times in a row”** — wait about 10 minutes and type the board’s current code.

Step 6: talk to Neddy

6.1 **Find the BOOT button.** Turn the board over in your hands and look at its **edge** for two tiny buttons. Next to each there is a tiny printed label: **BOOT** and **PWR**. Use **BOOT** for talking. (PWR is power: holding PWR for 6 seconds turns the board off; if that happens, press PWR once to turn it on again.)

If you cannot find the labels, send me a photo of the board's edge and I will mark which one it is.

6.2 Make sure the screen shows the green circle and **Hold BOOT to talk**. Be in a fairly quiet room and hold the board a hand's width from your mouth.

6.3 Press and **keep holding** BOOT. The screen turns to a **red circle** and says **Listening**, with a small bar near the bottom.

6.4 While holding it, speak clearly, for example: *"Hi Neddy, what can you do for me?"* The bar should move while you talk (that means the microphone hears you).

6.5 When you finish the sentence, **let go** of BOOT. Now the screen shows an **amber circle** and **Thinking**, then a teal circle with Neddy's words appearing as she speaks, and you **hear her** from the speaker.

6.6 When she is done the green screen comes back. Near the bottom it says something like **last answer 1.4 s after release**. That is how long she took to start answering after you let go. **Please tell me that number** each time you try; it is how we make her faster.

6.7 Try a second question straight away ("What's on my calendar today?"). You can also press BOOT while she is talking: she stops, like talking over a person.

Things you might notice, all normal:

- The very first question after a quiet spell can take a second or two longer: the call has to start.
- A very quick tap does nothing, on purpose. Hold for at least a second.
- One press can talk for up to 30 seconds.

What happens	What it means / what to do
Red circle, but the bar never moves	The microphone is not hearing you. Tell me, and send the Serial Monitor text (below).
Amber "Thinking" forever	The board got no reply. Wait 15 seconds, let go and try again; if it repeats, send the Serial Monitor text.

You see her words but hear nothing	Is a speaker attached to the board's speaker socket? Did the sound test in Part 1, step 20, play music? Tell me.
A message in words, like "Talking isn't switched on yet"	Read it to me exactly; it usually means something on the Neddy server needs turning on, not the board.
The board turns itself off	You held PWR, not BOOT. Press PWR once.

Sending me what the board says (the Serial Monitor)

★.1 With the board plugged in, in Arduino IDE click **Tools › Serial Monitor**. A panel opens at the bottom.

★.2 At the right of that panel, find the speed box and choose **115200 baud**.

★.3 Unplug the board and plug it back in. Lines appear, such as `[wifi] joined`, `[pair] show code`, `[neddy] connected`, `[talk] start`, `[latency] first sound 1380 ms after release`. Use the board for a minute.

★.4 Click in the panel, press `Ctrl` + `A` and `Ctrl` + `C`, and paste it into our chat. **Close the Serial Monitor** before pressing Upload again (it blocks the port).

Starting over: unplug the board, hold BOOT while plugging it in and keep holding for two seconds until the screen says **Pairing cleared**. You can also remove it under **Your devices** in the app. Then pair again from Step 5.